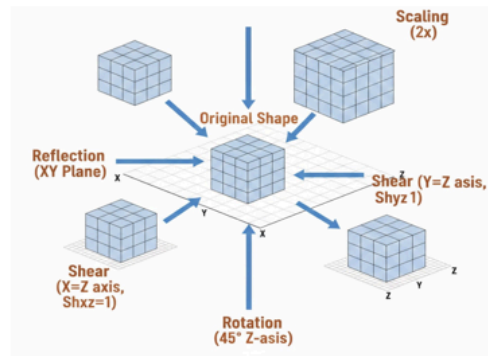


Computer Graphics

3D Transformation

Lecture 7



3D Transformations Types



The most common 3D transformations



Examples

Some examples of 3D transformations in computer graphics include:-

- Moving objects (3D Translation)
- Rotating objects around an axis (3D Rotation)
- Scaling objects (3D Scaling)
- Flipping objects (3D Reflection)
- Distorting objects along an axis (3D Shearing)
- Distorting objects along an axis (3D Shearing)

3D Transformations

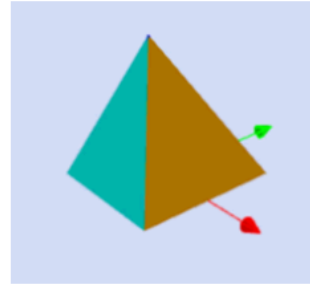
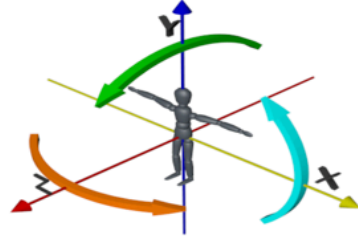


- When the transformation takes place on a 3D plane. It is called 3D transformation
- Generalize from 2D by including z coordinate Straight forward for translation and scale, rotation more difficult

Homogeneous coordinates: 4 components

Transformation matrices: 4×4 elements

$$\begin{bmatrix} a & b & c & t_x \\ d & e & f & t_y \\ g & h & i & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



3D Transformation — Lecture 7 & 11

সম্পূর্ণ বাংলায় বিস্তারিত নোট

আগে বুঝো — Transformation কী?

Transformation (রূপান্তর) মানে কোনো object-এর অবস্থান, আকার, দিক বা আকৃতি পরিবর্তন করা।

2D vs 3D Transformation:

বিষয়	2D	3D
Axis	X, Y	X, Y, Z
জটিলতা	সহজ	বেশি জটিল
ব্যবহার	সমতল ছবি	3D object/গেম/সিনেমা

3D-তে মোট ৫টি Transformation:

3D Transformation

- 1. Translation → সরানো
- 2. Rotation → ঘোরানো
- 3. Scaling → ছোট-বড় করা

- └─ 4. Reflection → আয়নায় প্রতিফলন
 - └─ 5. Shearing → বাঁকানো/তির্থক করা
-

1 Translation (ট্রান্সলেশন) — সরানো

সহজ সংজ্ঞা: 3D space-এ object-কে এক জায়গা থেকে অন্য জায়গায় সরানো।

মূল ধারণা:

- পুরোনো অবস্থান: **(Xold, Yold, Zold)**
- কতটুকু সরাবো: **Translation Vector = (Tx, Ty, Tz)**
- নতুন অবস্থান: **(Xnew, Ynew, Znew)**

$T_x \rightarrow$ X-অক্ষে কতটুকু সরাবো (বাম-ডান)

$T_y \rightarrow$ Y-অক্ষে কতটুকু সরাবো (উপর-নিচ)

$T_z \rightarrow$ Z-অক্ষে কতটুকু সরাবো (সামনে-পেছনে)

সূত্র:

$$X_{new} = X_{old} + T_x$$

$$Y_{new} = Y_{old} + T_y$$

$$Z_{new} = Z_{old} + T_z$$

 উদাহরণ: গেমের character সামনে এগিয়ে যাচ্ছে — এটা Translation। X, Y, Z তিন দিকেই সরতে পারে।

✓ Translation সমাধান (Problem 1):

দেওয়া: A(0,3,1), B(3,3,2), C(3,0,0), D(0,0,0) | Translation Vector = (1, 1, 2)

বিন্দু A(0, 3, 1):

$$X_{new} = 0 + 1 = 1$$

$$Y_{new} = 3 + 1 = 4$$

$$Z_{new} = 1 + 2 = 3 \rightarrow A' = (1, 4, 3)$$

বিন্দু B(3, 3, 2):

$$X_{new} = 3 + 1 = 4$$

$$Y_{new} = 3 + 1 = 4$$

$$Z_{new} = 2 + 2 = 4 \rightarrow B' = (4, 4, 4)$$

বিন্দু C(3, 0, 0):

$$X_{new} = 3 + 1 = 4$$

$$Y_{new} = 0 + 1 = 1$$

$$Z_{\text{new}} = 0+2 = 2 \rightarrow C' = (4, 1, 2)$$

বিন্দু D(0, 0, 0):

$$X_{\text{new}} = 0+1 = 1$$

$$Y_{\text{new}} = 0+1 = 1$$

$$Z_{\text{new}} = 0+2 = 2 \rightarrow D' = (1, 1, 2)$$

✓ উত্তর: A(1,4,3), B(4,4,4), C(4,1,2), D(1,1,2)

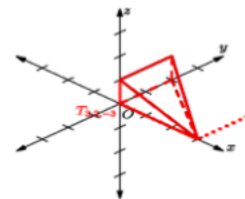
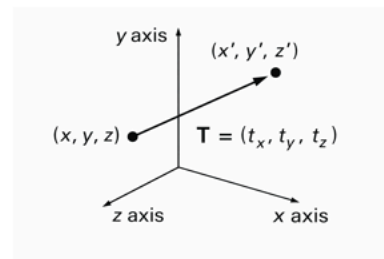
3D Transformations (Translation)



- Moving of an object is called translation.
- In 3-dimensional homogeneous coordinate representation, a point is transformed from position $P = (x, y, z)$ to $P' = (x', y', z')$
- This can be written as:- Using $P' = T \cdot P$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Translation of objects



3D Transformations (Translation)



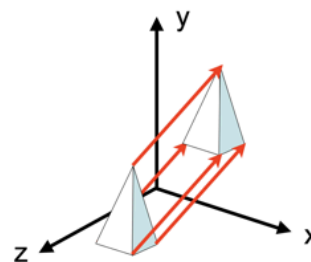
The matrix representation is equivalent to the three equation.

$$x' = x + t_x,$$

$$y' = y + t_y,$$

$$z' = z + t_z$$

Where parameter t_x, t_y, t_z are specifying translation distance for the coordinate directions x, y, and z, are assigned any real value.



2D Transformations (Translation Example)



Perform translation transformation on the following figure where the given translation distances are $t_x = 35$, $t_y = 25$, and $t_z = 10$

Solution:

Translation distances:

$$t_x = 35, \quad t_y = 25, \quad t_z = 10$$

3D translation formula

$$(x', y', z') = (x + t_x, y + t_y, z + t_z)$$

For Point A (10, 5, 8),

$$x' = 10 + 35 = 45$$

$$y' = 5 + 25 = 30$$

$$z' = 8 + 10 = 18$$

$$A'(45, 30, 18)$$

For Point B (30, 5, 10),

$$x' = 30 + 35 = 65$$

$$y' = 5 + 25 = 30$$

$$z' = 10 + 10 = 20$$

$$B'(65, 30, 20)$$

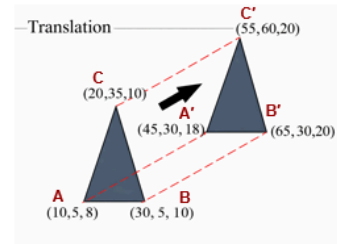
For Point C (20, 35, 10),

$$x' = 20 + 35 = 55$$

$$y' = 35 + 25 = 60$$

$$z' = 10 + 10 = 20$$

$$C'(55, 60, 20)$$



2 Rotation (রোটেশন) — ঘোরানো

সহজ সংজ্ঞা: 3D space-এ object-কে কোনো **axis**-কে কেন্দ্র করে নির্দিষ্ট কোণে ঘোরানো।

3D-তে তিনটি **axis**-এ ঘোরানো যায়:

Y
|
| X-axis ঘোরানো → সামনে-পেছনে কাত
| Y-axis ঘোরানো → বাম-ডান ঘোরানো
|—— X Z-axis ঘোরানো → সমতলে ঘোরানো
/
Z

🔄 উদাহরণ: পাথার পাখা Z-axis-এ ঘোরে। পৃথিবী Y-axis-এ ঘোরে। দরজা X-axis-এ খোলে।

গুরুত্বপূর্ণ মনে রাখো:

- θ (theta) = ঘোরানোর কোণ
- $\cos 90^\circ = 0$, $\sin 90^\circ = 1$ — exam-এ সবচেয়ে বেশি কাজে লাগে

Axis	অক্ষের দিক	ঘোরার ধরণ	মানবদেহের সহজ উদাহরণ
X-Axis	বাম ↔ ডান	সামনে-পেছনে (Pitch)	মাথা নেড়ে 'হ্যাঁ' বলা
Y-Axis	নিচ ↔ ওপর	বামে-ডানে (Yaw)	মাথা নেড়ে 'না' বলা
Z-Axis	সামনে ↔ পেছনে	ঘড়ির কাঁটার মতো (Roll)	কাঁধের দিকে মাথা কাত করা

● X-Axis Rotation:

X একই থাকে, Y ও Z বদলায়।

$$\begin{aligned} X_{\text{new}} &= X_{\text{old}} \\ Y_{\text{new}} &= Y_{\text{old}} \cos \theta - Z_{\text{old}} \sin \theta \\ Z_{\text{new}} &= Y_{\text{old}} \sin \theta + Z_{\text{old}} \cos \theta \end{aligned}$$

মনে রাখার কৌশল: X-এ ঘুরলে X অপরিবর্তিত, Y-Z বদলায়।

● Y-Axis Rotation:

Y একই থাকে, X ও Z বদলায়।

$$\begin{aligned} X_{\text{new}} &= Z_{\text{old}} \sin \theta + X_{\text{old}} \cos \theta \\ Y_{\text{new}} &= Y_{\text{old}} \\ Z_{\text{new}} &= Z_{\text{old}} \cos \theta - X_{\text{old}} \sin \theta \end{aligned}$$

● Z-Axis Rotation:

Z একই থাকে, X ও Y বদলায়।

$$\begin{aligned} X_{\text{new}} &= X_{\text{old}} \cos \theta - Y_{\text{old}} \sin \theta \\ Y_{\text{new}} &= X_{\text{old}} \sin \theta + Y_{\text{old}} \cos \theta \\ Z_{\text{new}} &= Z_{\text{old}} \end{aligned}$$

মনে রাখার কৌশল: যে axis-এ ঘোরাবে সেটা অপরিবর্তিত থাকে।

✓ Rotation সমাধান ($\theta = 90^\circ$):

দেওয়া: $(X_{old}, Y_{old}, Z_{old}) = (1, 2, 3)$, $\theta = 90^\circ$ $\cos 90^\circ = 0$, $\sin 90^\circ = 1$

X-Axis Rotation:

$$\begin{aligned} X_{new} &= 1 \\ Y_{new} &= 2 \times 0 - 3 \times 1 = -3 \\ Z_{new} &= 2 \times 1 + 3 \times 0 = 2 \\ &\rightarrow (1, -3, 2) \end{aligned}$$

Y-Axis Rotation:

$$\begin{aligned} X_{new} &= 3 \times 1 + 1 \times 0 = 3 \\ Y_{new} &= 2 \\ Z_{new} &= 3 \times 0 - 1 \times 1 = -1 \\ &\rightarrow (3, 2, -1) \end{aligned}$$

Z-Axis Rotation:

$$\begin{aligned} X_{new} &= 1 \times 0 - 2 \times 1 = -2 \\ Y_{new} &= 1 \times 1 + 2 \times 0 = 1 \\ Z_{new} &= 3 \\ &\rightarrow (-2, 1, 3) \end{aligned}$$

3D Transformations (Rotation)



■ X-AXIS ROTATION

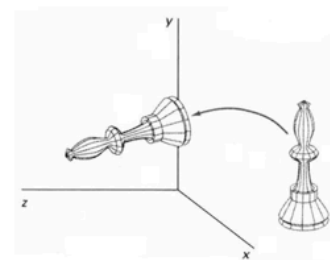
The equation for X-axis rotation

$$\mathbf{x}' = \mathbf{x}$$

$$y' = y \cos \theta - z \sin \theta$$

$$z' = y \sin \theta + z \cos \theta$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Transformations (Rotation)



Y-axis rotation:

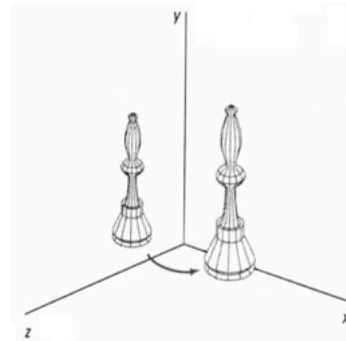
The equation for Y-axis rotation

$$x' = x \cos\theta + z \sin\theta$$

$$y' = y$$

$$z' = z \cos\theta - x \sin\theta$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



3D Transformations (Rotation)



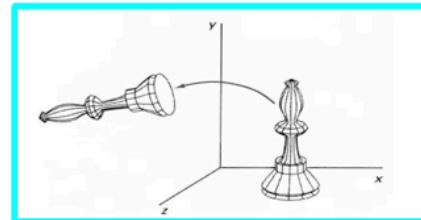
Z-axis rotation

The equation for Z-axis rotation

$$x' = x \cos\theta - y \sin\theta$$

$$y' = x \sin\theta + y \cos\theta$$

$$z' = z$$



$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

3D Transformations (Rotation Example)



Given a homogeneous point (1, 2, 3). Apply a 90-degree rotation towards the X, Y, and Z axis and find out the new coordinate points

Solution :

- Point: (x, y, z) = (1, 2, 3)
- Rotation angle $\theta = 90^\circ$

Rotation about X-axis

$$x' = x$$

$$y' = y \cos\theta - z \sin\theta$$

$$z' = y \sin\theta + z \cos\theta$$

$$x' = 1$$

$$y' = 2 \cos 90^\circ - 3 \sin 90^\circ = -3$$

$$z' = 2 \sin 90^\circ + 3 \cos 90^\circ = 2$$

New coordinate: (1, -3, 2)

Rotation about Y-axis

$$x' = z \sin\theta + x \cos\theta$$

$$y' = y$$

$$z' = z \cos\theta - x \sin\theta$$

$$x' = 3 \sin 90^\circ + 1 \cos 90^\circ = 3$$

$$y' = 2$$

$$z' = 3 \cos 90^\circ - 1 \sin 90^\circ = -1$$

New coordinate: (3, 2, -1)

3D Transformations (Rotation Example)



Given a homogeneous point (1, 2, 3). Apply a 90-degree rotation towards the X, Y, and Z axis and find out the new coordinate points

Solution :

- Point: (x, y, z) = (1, 2, 3)
- Rotation angle $\theta = 90^\circ$

Rotation about Z-axis

$$\begin{aligned}x' &= x \cos\theta - y \sin\theta \\y' &= x \sin\theta + y \cos\theta \\z' &= z\end{aligned}$$

$$\begin{aligned}x' &= 1 \cos 90^\circ - 2 \sin 90^\circ = -2 \\y' &= 1 \sin 90^\circ + 2 \cos 90^\circ = 1 \\z' &= 3\end{aligned}$$

New coordinate: (-2, 1, 3)

Final Results

X-axis rotation $\rightarrow (1, -3, 2)$

Y-axis rotation $\rightarrow (3, 2, -1)$

Z-axis rotation $\rightarrow (-2, 1, 3)$

3 Scaling (স্কেলিং) — আকার পরিবর্তন

সহজ সংজ্ঞা: object-কে বড় বা ছোট করা।

Scaling Factor (স্কেলিং ফ্যাক্টর):

- $S_x > 1 \rightarrow X$ দিকে বড় হবে
- $S_x < 1 \rightarrow X$ দিকে ছোট হবে
- $S_x = 1 \rightarrow$ অপরিবর্তিত

সূত্র:

$$X_{\text{new}} = X_{\text{old}} \times S_x$$

$$Y_{\text{new}} = Y_{\text{old}} \times S_y$$

$$Z_{\text{new}} = Z_{\text{old}} \times S_z$$

📱 উদাহরণ: ফোনে ছবি zoom করলে — Scaling হচ্ছে। $S_x=S_y=S_z=2$ মানে সব দিকে দ্বিগুণ।

✓ Scaling সমাধান:

দেওয়া: A(0,3,3), B(3,3,6), C(3,0,1), D(0,0,0) | $S_x=2, S_y=3, S_z=3$

A(0, 3, 3):

$$0 \times 2 = 0, 3 \times 3 = 9, 3 \times 3 = 9 \rightarrow A' = (0, 9, 9)$$

$$B(3, 3, 6):$$

$$3 \times 2 = 6, 3 \times 3 = 9, 6 \times 3 = 18 \rightarrow B' = (6, 9, 18)$$

$$C(3, 0, 1):$$

$$3 \times 2 = 6, 0 \times 3 = 0, 1 \times 3 = 3 \rightarrow C' = (6, 0, 3)$$

$$D(0, 0, 0):$$

$$0, 0, 0 \rightarrow D' = (0, 0, 0)$$

3D Transformations (Scaling)



- Changes the size of the object and repositions the object relative to the coordinate origin.

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

The equations for scaling,

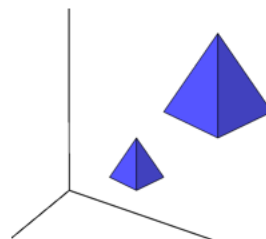
$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

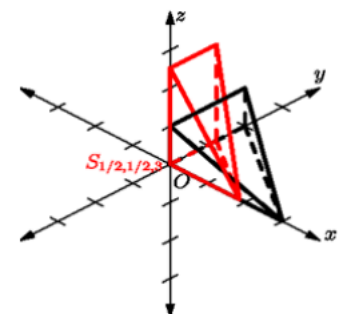
$$z' = S_z \cdot z$$

Where:

- S_x = scaling factor along X-axis
- S_y = scaling factor along Y-axis
- S_z = scaling factor along Z-axis



Scaling of objects



3D Transformations (Scaling Example)



Consider the above problem where a cube "OABCDEFG" is given with O(0, 0, 0), A(0, 4, 0), B(0, 4, 4), C(4, 4, 0), D(4, 4, 4), E(4, 0, 0), F(0, 0, 4), and G(4, 0, 4) and we are given the scaling factors S_x , S_y , and S_z . Perform a scaling operation over the cube.

Scaling factors: $S_x = 2$, $S_y = 3$, $S_z = 2$

Transformation Equation,

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P' = S \times P$$

For a point P(x,y,z),

$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

$$z' = S_z \cdot z$$

3D Transformations (Scaling Example)



Consider the above problem where a cube "OABCDEFG" is given with O(0, 0, 0), A(0, 4, 0), B(0, 4, 4), C(4, 4, 0), D(4, 4, 4), E(4, 0, 0), F(0, 0, 4), and G(4, 0, 4) and we are given the scaling factors S_x , S_y , and S_z . Perform a scaling operation over the cube.

For point O (0,0,0)

$$x' = 2 \times x = 2 \times 0 = 0$$

$$y' = 3 \times y = 3 \times 0 = 0$$

$$z' = 2 \times z = 2 \times 0 = 0$$

Scaled Point O' (0, 0, 0)

For point A (0, 4, 0)

$$x' = 2 \times x = 2 \times 0 = 0$$

$$y' = 3 \times y = 3 \times 4 = 12$$

$$z' = 2 \times z = 2 \times 0 = 0$$

Scaled Point A' (0, 12, 0)

For point B (0, 4, 4)

$$x' = 2 \times x = 2 \times 0 = 0$$

$$y' = 3 \times y = 3 \times 4 = 12$$

$$z' = 2 \times z = 2 \times 4 = 8$$

Scaled Point B' (0, 12, 8)

For point C (4, 4, 0)

$$x' = 2 \times x = 2 \times 4 = 8$$

$$y' = 3 \times y = 3 \times 4 = 12$$

$$z' = 2 \times z = 2 \times 0 = 0$$

Scaled Point C' (8,12,0)

For point D (4, 4, 4)

$$x' = 2 \times x = 2 \times 4 = 8$$

$$y' = 3 \times y = 3 \times 4 = 12$$

$$z' = 2 \times z = 2 \times 4 = 8$$

Scaled Point D' (8, 12, 8)

For Point E (4, 0, 0)

$$x' = 2 \times x = 2 \times 4 = 8$$

$$y' = 3 \times y = 3 \times 0 = 0$$

$$z' = 2 \times z = 2 \times 0 = 0$$

Scaled Point E' (8,0,0)

3D Transformations (Scaling Example)



Consider the above problem where a cube "OABCDEFG" is given with $O(0, 0, 0)$, $A(0, 4, 0)$, $B(0, 4, 4)$, $C(4, 4, 0)$, $D(4, 4, 4)$, $E(4, 0, 0)$, $F(0, 0, 4)$, and $G(4, 0, 4)$ and we are given the scaling factors S_x , S_y , and S_z . Perform a scaling operation over the cube.

For point F (0,0,4)

$$x' = 2 \times x = 2 \times 0 = 0$$

$$y' = 3 \times y = 3 \times 0 = 0$$

$$z' = 2 \times z = 2 \times 4 = 8$$

Scaled Point F' (0, 0, 8)

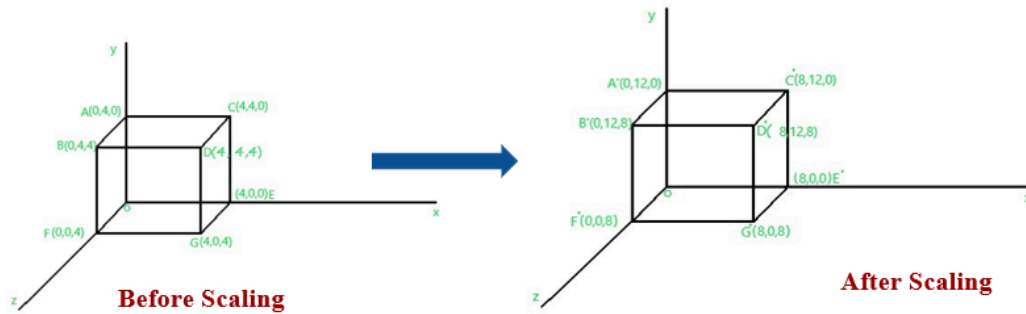
For point G (4, 0, 4)

$$x' = 2 \times x = 2 \times 4 = 8$$

$$y' = 3 \times y = 3 \times 0 = 0$$

$$z' = 2 \times z = 2 \times 4 = 8$$

Scaled Point A' (8, 0, 8)



4 Reflection (রিফ্লেকশন) — আয়নার প্রতিফলন

সহজ সংজ্ঞা: object-এর আয়না-প্রতিচ্ছবি তৈরি করা। এটা আসলে **180° rotation**।

বৈশিষ্ট্য:

- আকার একই থাকে, শুধু অবস্থান পরিবর্তন
- সবসময় কোনো **plane**-এর অপর পাশে প্রতিফলন হয়

3D-তে তিনটি plane-এ reflection হয়:

● XY Plane-এ Reflection:

XY plane হলো মেঝের মতো সমতল। Z উল্টো হয়।

$$X_{\text{new}} = X_{\text{old}}$$

$$Y_{\text{new}} = Y_{\text{old}}$$

$$Z_{\text{new}} = -Z_{\text{old}} \quad \leftarrow \text{শুধু } Z \text{ উল্টো}$$

উদাহরণ: মেঝেতে পানি — তার মধ্যে উপরের জিনিসের প্রতিফলন। উপরের Z নিচে (-Z) চলে যায়।

● YZ Plane-এ Reflection:

YZ plane হলো দেয়ালের মতো। X উল্টো হয়।

$$X_{\text{new}} = -X_{\text{old}} \quad \leftarrow \text{শুধু X উল্টো}$$

$$Y_{\text{new}} = Y_{\text{old}}$$

$$Z_{\text{new}} = Z_{\text{old}}$$

● XZ Plane-এ Reflection:

XZ plane হলো মাটির মতো। Y উল্টো হয়।

$$X_{\text{new}} = X_{\text{old}}$$

$$Y_{\text{new}} = -Y_{\text{old}} \quad \leftarrow \text{শুধু Y উল্টো}$$

$$Z_{\text{new}} = Z_{\text{old}}$$

মনে রাখার সহজ নিয়ম:

যে plane-এ reflection হয়, সেই plane-এর বাইরের **axis** উল্টো (-) হয়। XY plane $\rightarrow$ Z বাদ, তাই **Z** উল্টো YZ plane $\rightarrow$ X বাদ, তাই **X** উল্টো XZ plane $\rightarrow$ Y বাদ, তাই **Y** উল্টো

✓ Reflection সমাধান (XY Plane):

দেওয়া: A(3,4,1), B(6,4,2), C(5,6,3)

$$A(3, 4, 1) \rightarrow (3, 4, -1)$$

$$B(6, 4, 2) \rightarrow (6, 4, -2)$$

$$C(5, 6, 3) \rightarrow (5, 6, -3)$$

✓ Reflection সমাধান (XZ Plane):

$$A(3, 4, 1) \rightarrow (3, -4, 1)$$

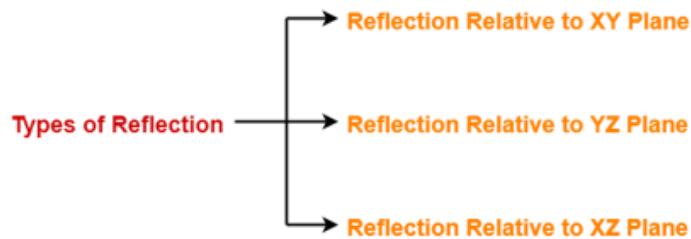
$$B(6, 4, 2) \rightarrow (6, -4, 2)$$

$$C(5, 6, 3) \rightarrow (5, -6, 3)$$

3D Transformations (Reflection)



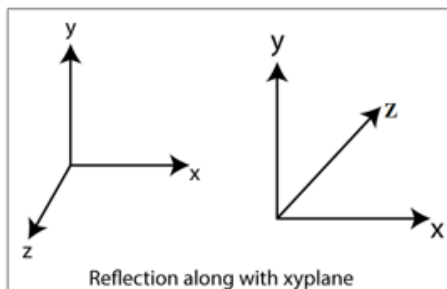
- Reflection in computer graphics is used to emulate reflective objects like mirrors and shiny surfaces. Reflection may be an x-axis, y-axis, z-axis, and also in the planes xy-plane, yz-plane, and zx-plane.
- Reflection relative to a given Axis are equivalent to 180-degree rotations



3D Transformations (Reflection)



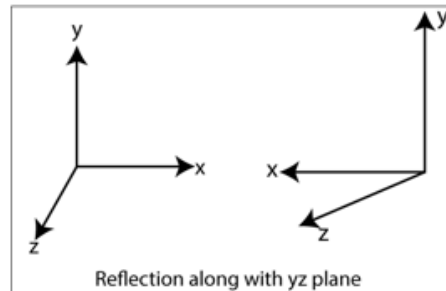
Reflection Relative to XY Plane:



This reflection is achieved by using the following reflection equations:

$$\begin{aligned}x' &= x \\y' &= y \\z' &= -z\end{aligned}$$

Reflection Relative to YZ Plane:



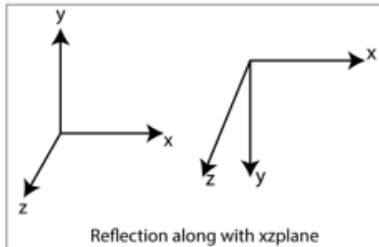
This reflection is achieved by using the following reflection equations:

$$\begin{aligned}x' &= -x \\y' &= y \\z' &= z\end{aligned}$$

3D Transformations (Reflection)



Reflection Relative to XZ Plane:



This reflection is achieved by using the following reflection equations:

$$x' = x$$

$$y' = -y$$

$$z' = z$$

Reflection along with xzplane

3D Transformations (Reflection)



Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), and C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object.

Solution:

Vertices of triangle:

A(3, 4, 1)

B(6, 4, 2)

C(5, 6, 3)

Reflection plane: XY-plane

$$x' = x$$

$$y' = y$$

$$z' = -z$$

Only z-coordinate changes sign

For point A (3, 4, 1)

$$x' = 3$$

$$y' = 4$$

$$z' = -1$$

New coordinate, A' = (3, 4, -1)

For point B(6, 4, 2)

$$x' = 6$$

$$y' = 4$$

$$z' = -2$$

New coordinate, B' = (6, 4, -2)

For point C(5, 6, 3)

$$x' = 5$$

$$y' = 6$$

$$z' = -3$$

New coordinate, C' = (5, 6, -3)

Final Results: A' (3, 4, -1), B' (6, 4, -2), C' (5, 6, -3)

5 Shearing (শিয়ারিং) — বাঁকানো

সহজ সংজ্ঞা: object-এর আকৃতি পরিবর্তন করা — একটা দিক ঠিক রেখে অন্য দিক টেনে বাঁকানো।

Shearing Parameter:

- $Sh_x \rightarrow X$ দিকের shear
- $Sh_y \rightarrow Y$ দিকের shear
- $Sh_z \rightarrow Z$ দিকের shear

 উদাহরণ: একটা চারকোণা বাক্সকে ধাক্কা দিলে যদি সমান্তরিক চতুর্ভুজ হয়ে যায় — এটাই Shearing। একটা দিক স্থির, অন্যটা সরে যায়।

তিনটো **axis-এ Shearing** সূত্র:

X-Axis Shearing (X অপরিবর্তিত, Y ও Z বদলায়):

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} \\Y_{\text{new}} &= Y_{\text{old}} + Sh_y \times X_{\text{old}} \\Z_{\text{new}} &= Z_{\text{old}} + Sh_z \times X_{\text{old}}\end{aligned}$$

Y-Axis Shearing (Y অপরিবর্তিত, X ও Z বদলায়):

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} + Sh_x \times Y_{\text{old}} \\Y_{\text{new}} &= Y_{\text{old}} \\Z_{\text{new}} &= Z_{\text{old}} + Sh_z \times Y_{\text{old}}\end{aligned}$$

Z-Axis Shearing (Z অপরিবর্তিত, X ও Y বদলায়):

$$\begin{aligned}X_{\text{new}} &= X_{\text{old}} + Sh_x \times Z_{\text{old}} \\Y_{\text{new}} &= Y_{\text{old}} + Sh_y \times Z_{\text{old}} \\Z_{\text{new}} &= Z_{\text{old}}\end{aligned}$$

মনে রাখার নিয়ম: যে axis-এ shear হচ্ছে সেটা অপরিবর্তিত, বাকি দুটো বদলায়।

 **Shearing** সমাধান:

দেওয়া: A(0,0,0), B(1,1,2), C(1,1,3) | $Sh_x=2$, $Sh_y=2$, $Sh_z=3$

X-Axis Shearing:

$$\begin{aligned}A(0,0,0): & 0, 0+2\times0=0, 0+3\times0=0 \rightarrow (0, 0, 0) \\B(1,1,2): & 1, 1+2\times1=3, 2+3\times1=5 \rightarrow (1, 3, 5) \\C(1,1,3): & 1, 1+2\times1=3, 3+3\times1=6 \rightarrow (1, 3, 6)\end{aligned}$$

Y-Axis Shearing:

$$\begin{aligned}A(0,0,0): & 0+2\times0=0, 0, 0+3\times0=0 \rightarrow (0, 0, 0) \\B(1,1,2): & 1+2\times1=3, 1, 2+3\times1=5 \rightarrow (3, 1, 5) \\C(1,1,3): & 1+2\times1=3, 1, 3+3\times1=6 \rightarrow (3, 1, 6)\end{aligned}$$

Z-Axis Shearing:

A(0,0,0): $0+2\times0=0$, $0+2\times0=0$, $0 \rightarrow (0, 0, 0)$

B(1,1,2): $1+2\times2=5$, $1+2\times2=5$, $2 \rightarrow (5, 5, 2)$

C(1,1,3): $1+2\times3=7$, $1+2\times3=7$, $3 \rightarrow (7, 7, 3)$

সব সূত্র এক জায়গায় — Quick Reference

Transformation	সূত্র	যা অপরিবর্তিত
Translation	$X+Tx, Y+Ty, Z+Tz$	—
X Rotation	$X, Y\cdot\cos\theta-Z\cdot\sin\theta, Y\cdot\sin\theta+Z\cdot\cos\theta$	X
Y Rotation	$Z\cdot\sin\theta+X\cdot\cos\theta, Y, Z\cdot\cos\theta-X\cdot\sin\theta$	Y
Z Rotation	$X\cdot\cos\theta-Y\cdot\sin\theta, X\cdot\sin\theta+Y\cdot\cos\theta, Z$	Z
Scaling	$X\cdot Sx, Y\cdot Sy, Z\cdot Sz$	—
XY Reflection	$X, Y, -Z$	X, Y
YZ Reflection	$-X, Y, Z$	Y, Z
XZ Reflection	$X, -Y, Z$	X, Z
X Shearing	$X, Y+Shy\cdot X, Z+Shz\cdot X$	X
Y Shearing	$X+Shx\cdot Y, Y, Z+Shz\cdot Y$	Y
Z Shearing	$X+Shx\cdot Z, Y+Shy\cdot Z, Z$	Z

সব মনে রাখার সহজ কৌশল

Rotation ও Shearing-এ: যে axis-এ কাজ হচ্ছে সেই axis অপরিবর্তিত থাকে।

Reflection-এ: যে plane-এ প্রতিফলন, সেই plane-এর বাইরের **axis** উল্টো (-) হয়।

Scaling-এ: >1 মানে বড়, <1 মানে ছোট।

Translation-এ: শুধু যোগ করো।